

ASTER: Latent Pseudo-Anomaly Generation for Unsupervised Time-Series Anomaly Detection

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Abstract. Time-series anomaly detection (TSAD) is critical in domains such as industrial monitoring, healthcare, and cybersecurity, but it remains challenging due to rare and heterogeneous anomalies and the scarcity of labelled data. This scarcity makes unsupervised approaches predominant, yet existing methods often rely on reconstruction or forecasting, which struggle with complex data, or on embedding-based approaches that require domain-specific anomaly synthesis and fixed distance metrics. We propose ASTER, a framework that generates pseudo-anomalies directly in the latent space, avoiding handcrafted anomaly injections and the need for domain expertise. A latent-space decoder produces tailored pseudo-anomalies to train a Transformer-based anomaly classifier, while a pre-trained LLM enriches the temporal and contextual representations of this space. Experiments on three benchmark datasets show that ASTER achieves state-of-the-art performance and sets a new standard for LLM-based TSAD.

Keywords: Anomaly Detection · Time-Series · Pseudo-Anomalies · LLM.

1 Introduction

Time-series anomaly detection (TSAD) focuses on identifying patterns or behaviours in temporal data that deviate from expected dynamics. It plays a critical role in a wide range of real-world applications, including industrial fault diagnosis [45], financial fraud detection [13], healthcare monitoring [23], and cybersecurity [19]. Unlike anomaly detection on static supports, such as images or 3D data [26,51,25], TSAD must account for temporal dependencies between observations and evolving data distributions, making the task considerably more complex. Specifically, detecting anomalies in time-series is inherently challenging due to the rarity of anomalous events, their heterogeneous temporal patterns, and their dependence on contextual information, requiring models to account for both temporal dynamics and interactions across multiple features.

<https://gitlab.com/uniluxembourg/snt/cvi2/open/space/aster-tab>

These challenges are compounded by the scarcity of labelled anomalies in practical scenarios, motivating the development of unsupervised approaches [57,34] that can effectively model normal behaviour and detect deviations without reliance on annotated data.

Among unsupervised methods, we find embedding-based strategies [34,30]. They aim to learn a transformation that maps normal patterns to a dense region in the representation space, effectively separating them from abnormal patterns. In this paradigm, we commonly find approaches that aim to maximise the distance between the embeddings of normal data and synthetically generated anomalies—or *pseudo-anomalies*. For these methods, the accurate and realistic generation of synthetic anomalies is crucial for training a reliable anomaly detection model. To this end, most rely on domain knowledge to tediously curate lists of possible anomaly types. These selective lists are then used as a basis for defining transformations, such as adding bias, noise, or masking. These corruptive transformations can then be applied to normal data to generate pseudo-anomalies. However, considering the nature of the multi-variate TSAD problem, this procedural, data augmentation-dependent anomaly generation is confined to the curated lists and restricts the performance and generalisation of the models.

An alternative direction could involve making the transformations adapt automatically to the data by *learning* them; and to further improve generalisation, applying them within a less domain-specific space, *i.e.*, a latent representation space. This, however, requires the ability to construct expressive latent representations that capture temporal and cross-feature dynamics. Unfortunately, robust foundational models for time-series remain scarce, owing to data heterogeneity and limited availability [9]. Recently, Large Language Models (LLMs) have been adapted to solve various time-series problems, such as forecasting [29], classification [69], and anomaly detection [56]. These pre-trained models have proven to transfer helpful cross-domain and cross-modality knowledge thanks to their strong representation capabilities [44,53]. Additionally, given the rapid advancements and expanding capabilities of LLMs, formulating a TSAD method adapted to their architecture represents a valuable research opportunity.

In this work, we introduce ASTER, *Anomaly Synthesis through Transformer-based Embedding Representation*, a novel framework for unsupervised TSAD. ASTER generates pseudo-anomalies directly in latent space via learnt transformations, avoiding handcrafted, domain-specific augmentations in the raw data space. A pre-trained LLM is first used to encode time-series windows into rich, contextualised embeddings. These embeddings are then processed by a VAE-based module, the *perturbator*, which synthesises diverse and non-trivial pseudo-anomalies. Unlike metric-based approaches that depend on fixed distance measures, ASTER trains a Transformer-based classifier to learn a flexible decision boundary. This design enables ASTER to capture complex abnormal behaviours without relying on predefined anomaly types.

Our results show that ASTER achieves a new state-of-the-art performance on multiple datasets and improves upon the previous best LLM-based methods. We summarise our contributions as follows:

- We propose an unsupervised TSAD framework that generates pseudo-anomalies in latent space, eliminating domain-specific augmentations and improving generalisation.
- We introduce ASTER, a novel architecture featuring a VAE-based *perturbator* that models normal data and learns to generate pseudo-anomalies. It is paired with a Transformer-based classifier that learns a discriminative decision boundary.
- We demonstrate how pre-trained LLMs can be effectively leveraged as contextual feature extractors for TSAD.
- We validate our method using the rigorous and unified TAB benchmark [48]. This addresses inconsistencies in prior evaluations by enabling fair and reproducible comparisons across diverse TSAD methods.

The remainder of the paper is organised as follows: we provide an extensive review of related works in Section 2. Next, we formally introduce the problem and the proposed method in Section 3 and Section 4, respectively. Our results and ablation studies are presented in Section 5. We then discuss certain limitations and directions for future work in Section 6, and conclude in Section 7.

2 Related Work

In this section, we first provide a summary of TSAD methods, then focus on methods that actively refine embeddings of time-series data to reflect temporal patterns and elaborate on the adoption of LLMs for various time-series problems. Finally, we explain the efforts to provide a comprehensive benchmark for a fair evaluation and comparison of TSAD methods.

2.1 Time-Series Anomaly Detection

Early TSAD research relied on classical statistical techniques [65] and traditional machine learning methods, including autoregressive models, hypothesis testing, and distance-, density-, or isolation-based approaches [10,8,38]. However, these methods often struggle with complex temporal dependencies and non-linear patterns in high-dimensional data.

Deep learning has enabled models that learn expressive representations from raw inputs [51]. LSTM-based models [28], VAEs [63], and diffusion-based methods [12] improve reconstruction of normal patterns, while Generative Adversarial Networks (GANs) [66,68] introduce adversarial anomaly synthesis, and one-class classifiers like Deep SVDD [51] enhance robustness to noisy pseudo-labels.

Unsupervised paradigms dominate due to a scarcity of labelled anomalies. Reconstruction-, forecasting-, and imputation-based [4,55,61] methods detect anomalies through deviations from expected patterns. Self-supervised techniques, such as contrastive learning, masked modelling, or transformation prediction [64], further improve latent representations. Transformer-based models [57,42,58] capture long-range dependencies, improving performance on multivariate time-series.

2.2 Embedding-based TSAD

Shaping appropriate data embeddings suggests extracting informative features that capture the underlying data structures without task-specific supervision [5,35]. This is, in fact, closely related to the field of representation learning [64].

Contrastive learning [46,11] has emerged as a powerful paradigm for unsupervised TSAD [15]. By maximising alignment between augmented views of a sample (or similar ones) while distancing others, these models learn discriminative embeddings without labels. Contrastive TSAD methods, such as CTAD [30] and CATS [34], rely on specific augmentations (*e.g.*, masking, additive bias) to generate synthetic anomalies as negative samples, maximising their distance from normal representations using the contrastive loss. However, these strategies often require dataset-specific hyper-parameter tuning, limiting generalisation. CARLA [15] partially addresses this by confining the augmentation-based generation to a pre-training stage. Despite their success, these methods remain limited by manual, domain-intensive augmentation design in raw input space.

Latent space exploration, on the other hand, has traditionally focused on modelling normal data distributions. A prominent example is DAGMM [71,6], which utilises Gaussian Mixture Models (GMMs) in the latent space to identify low-probability deviations as anomalies. Similarly, TimeVQVAE-AD [36] employs a two-stage time-frequency auto-encoder to detect anomalies based on estimated densities and reconstruction quality. An increasing interest in latent space manipulation is also noticeable; for instance, L-GTA [50] applies augmentations directly in the latent space to enhance reconstruction robustness.

Departing from these approaches, our method does not seek to improve reconstruction performance, nor do we rely on passive density modelling of normal data. Instead, we leverage the latent space to model a specific distribution for pseudo-anomalies. By sampling directly from this learnt latent distribution, we circumvent the need for heuristic-driven augmentations in the raw input space, which are often suboptimal and domain-dependent. This strategy, concurrently explored in GenIAS [14], facilitates a more versatile, dataset-agnostic representation that removes the requirement for extensive domain knowledge. Consequently, the latent space and pseudo-anomaly generation process are optimised for the downstream task of anomaly detection.

2.3 LLMs & Time-Series

Recently, Large Language Models (LLMs) [49,17], which have excelled at capturing sequential relations in text data, have been adopted in the time-series domain to solve various tasks, such as forecasting [24,9,55,54,29], reasoning [62], and anomaly detection [69,37,2,56,39,70].

GPT4TS [69] is an early work that adapts a pre-trained GPT-2 [49] model for time-series tasks, including anomaly detection, using lightweight fine-tuning. Following this direction, UniTime [41] introduces masking-based input processing and incorporates domain knowledge via textual descriptions. AnomalyLLM [37] adopts a knowledge distillation framework, where a Transformer-based student

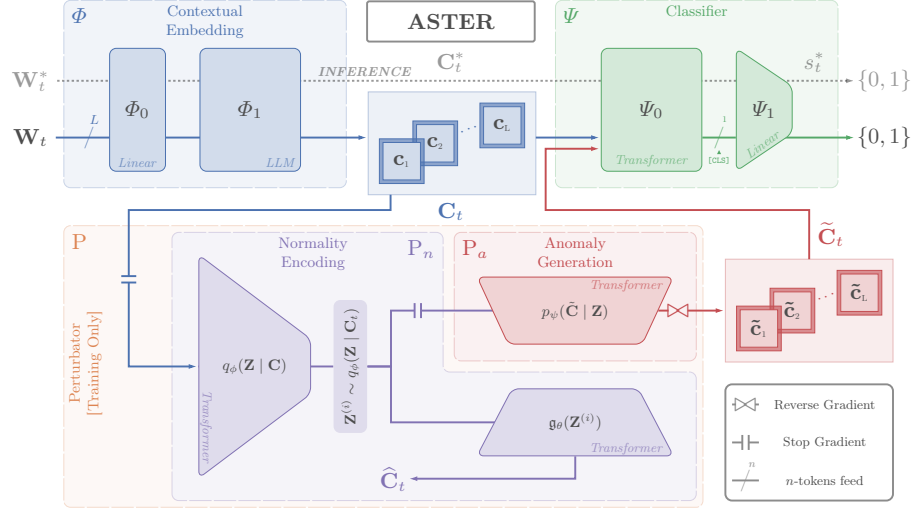


Fig. 1: Overview of the proposed method. It consists of three components: Contextual Embedding (Φ) extractor, Perturbator (P), and Classifier (Ψ). The Contextual Embedding maps the input time series to a high-dimensional space using a pre-trained LLM. The Perturbator learns a latent representation via reconstruction (P_n) and generates pseudo-anomalies (P_a). The Classifier distinguishes between normal and pseudo-anomalous data.

model learns from a GPT-2-initialised teacher, leveraging reconstruction errors for anomaly detection.

Another line of work explores the zero-shot capabilities of LLMs for time-series anomaly detection. SIGLLM [2] formulates anomaly detection as either direct prompting or forecasting-based discrepancy estimation. LLMAD [39] employs retrieval-augmented in-context learning to mimic expert reasoning, while AnomLLM [70] systematically analyses common assumptions in LLM-based TSAD, showing that longer input windows and Chain-of-Thought prompting do not consistently improve performance.

Motivated by these findings, we leverage a pre-trained LLM as a contextual feature extractor with minimal fine-tuning to bridge domain gaps, without introducing additional textual prompts or expert-defined context.

2.4 Benchmarking TSAD Models

Fair benchmarking of TSAD methods is crucial for accurately assessing progress. Methods achieving state-of-the-art performance under specific protocols or metrics, *e.g.*, point-adjusted F_1 -score, often degrade under more realistic criteria such as the standard F_1 -score. Several benchmarking efforts have been proposed for univariate and multivariate TSAD [67,7], but they largely exclude emerging paradigms, including LLM-based approaches and pre-training-based foundation

models. Recently, TAB [48] introduced a unified benchmark that standardises preprocessing, training, and evaluation, enabling fair and reproducible comparisons. We therefore adopt TAB and its baselines for all evaluations.

3 Problem Formulation

Let $\mathcal{X}^0 := \{\mathbf{x}_t \in \mathbb{R}^D\}_{t=1}^T$ be a multivariate time-series of length T , with corresponding labels $\mathcal{Y}^0 := \{y_t \in \{0, 1\}\}_{t=1}^T$, indicating normal and anomalous observations. The goal of TSAD is to learn a function f that assigns an anomaly score $s_t = f(\cdot)$ to each observation. Since anomalies are context-dependent, f operates on a window of the past L observations, *i.e.*, $s_t = f([\mathbf{x}_{t-L+1}, \dots, \mathbf{x}_t])$, where $f : \mathbb{R}^{L \times D} \rightarrow \mathbb{R}$. We thus process the time-series as a set of overlapping *windows* $\mathcal{W}^0 := \{\mathbf{W}_t\}_{t=L}^T$, with $\mathbf{W}_t := [\mathbf{x}_{t-L+1}, \dots, \mathbf{x}_t]$.

At inference, an unseen time series $\mathcal{X}^* := \{\mathbf{x}_t\}_{t=1}^{T'}$, drawn from the same domain as $\mathcal{X}^0$, is used for evaluation. Anomaly scores are computed sequentially for each window in $\mathcal{W}^*$, yielding $\mathcal{S}^* := \{s_t^*\}_{t=L}^{T'}$. A decision threshold $\tau \in \mathbb{R}$ is then applied to obtain predicted labels $\hat{\mathcal{Y}}^* := \{\hat{y}_t^*\}_{t=L}^{T'}$, where $\hat{y}_t^* = \mathbb{I}[s_t^* \geq \tau] \in \{0, 1\}$ indicates an anomalous time-step. Finally, comparing $\hat{\mathcal{Y}}^*$ with the ground-truth labels $\mathcal{Y}^*$ provides the model’s performance on the unseen series.

The task is challenging for two reasons: anomaly detection is context-dependent, and the setting is unsupervised ($y_t = 0, \forall t$), making learning f non-trivial. To prevent degenerate solutions, we generate a pseudo-anomalous window $\widehat{\mathbf{W}}_t$ for each input $\mathbf{W}_t$, with the challenge being to create meaningful windows.

4 Methodology

We present our architecture in Figure 1. The final objective f combines a contextual embedding $\Phi : \mathbb{R}^{L \times D} \rightarrow \mathbb{R}^{L \times M}$ (Section 4.1) and a classifier $\Psi : \mathbb{R}^{L \times M} \rightarrow \mathbb{R}$ (Section 4.2), so that for a previously unseen window $\mathbf{W}_t^*$ the anomaly score is $s_t^* = \Psi \circ \Phi(\mathbf{W}_t^*)$. To train Ψ effectively, we introduce a pseudo-anomaly generator, the *perturbator* $P : \mathbb{R}^{L \times M} \rightarrow \mathbb{R}^{L \times M}$, which produces plausible artificial anomalies (Section 4.3).

4.1 Time-Series to Contextualised Embeddings

Anomalies in time-series are highly context-dependent, motivating the use of LLMs to capture rich sequential patterns. This is the role of Φ , which we divide into two parts: $\Phi_0 : \mathbb{R}^D \rightarrow \mathbb{R}^M$, a time-step embedding, and $\Phi_1 : \mathbb{R}^{L \times M} \rightarrow \mathbb{R}^{L \times M}$, which adds sequential context.

Translation. Observations are mapped from the raw time-series to the token space via a linear projection, which has been shown effective in prior works [69, 40, 52]. This step only embeds individual time-steps, not cross-temporal knowledge:

$$\Phi_0 : \mathbb{R}^D \rightarrow \mathbb{R}^M \quad \Phi_0(\mathbf{W}_t^{[l]}), \quad l \in t - L + 1 \dots t \quad (1)$$

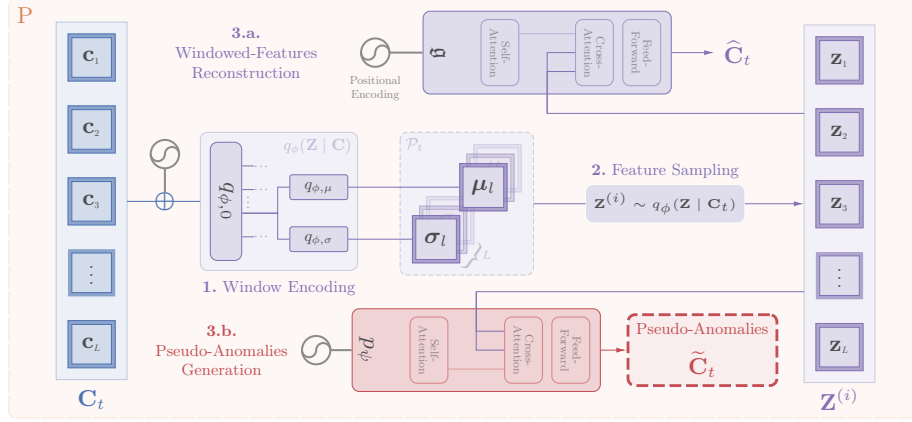


Fig. 2: Detailed description of the Perturbator architecture. It includes a complete Variational Auto-Encoder (purple), with an encoder q_ϕ that maps the input features $\mathbf{C}_t$ to latent matrix $\mathbf{Z}^{(i)}$, and a decoder g_θ . The pseudo-anomalies are generated using the additional decoder p_ψ (red), starting from the same latent matrix. We also use positional encoding added to the inputs of each sub-network.

LLM Knowledge. The LLM incorporates the entire window with prior sequential knowledge, yielding a contextualised window $\mathbf{C}_t \in \mathbb{R}^{L \times M}$ when combined with the translation module (Equation 2). Φ_0 is trained from scratch, while Φ_1 is only fine-tuned with LoRA [27], using gradients from the classification loss on normal windows $\nabla_\phi = \frac{\partial \mathcal{L}_{CE}}{\partial \mathbf{C}_t}$. This keeps part of the network independent of pseudo-anomaly generation, reducing the risk of collapse or overfitting.

$$\mathbf{C}_t = \Phi(\mathbf{W}_t) = \Phi_1 \circ \left(\left[\Phi_0(\mathbf{W}_j^{[l]}) \right]_{l=t-L+1}^t \right). \quad (2)$$

4.2 Classifier

For the classifier Ψ , we follow a BERT-like strategy [17]. The contextualised tokens $\mathbf{C}_t$ are first processed by $\Psi_0 : \mathbb{R}^{L \times M} \rightarrow \mathbb{R}^{L \times M}$ (Transformer-based network) to gather contextual information into a [CLS] token. This token is selected using $\Pi_1(\mathbf{X}) := [\mathbf{X}]_{:,1}$ and projected to a final score by $\Psi_1 : \mathbb{R}^M \rightarrow \mathbb{R}$:

$$\Psi := \Psi_1 \circ \Pi_1 \circ \Psi_0, \quad s_t = \Psi(\mathbf{C}_t). \quad (3)$$

Applying a sigmoid, $\bar{s}_t = \frac{1}{1+e^{-s_t}}$, allows optimisation with the binary cross-entropy loss (Equation 4). Together with the contextual embedding Φ , Ψ completes the required components for the inference pipeline of ASTER.

$$\mathcal{L}_{CE} = -[y_t \log(\bar{s}_t) + (1 - y_t) \log(1 - \bar{s}_t)]. \quad (4)$$

4.3 Perturbator

As mentioned in Section 4, during training we use a perturbator P to generate pseudo-anomalous samples and balance the data. We establish three essential requirements for the perturbator:

1. *Adaptive generation.* The process should automatically capture the structure of the data. Current methods rely on fixed transformations (*e.g.*, noise, shuffling, masking), which perform inconsistently across heterogeneous datasets. A robust method must generalise without dataset-specific tuning;

2. *Non-trivial anomalies.* While a simple alteration, such as adding random noise, suffices to simulate anomalies, these pseudo-anomalies are obvious when compared to the subtle, context-dependent nature of real anomalies. Pseudo-anomalies should remain correlated with normal data while being hard to detect;

3. *Diverse anomalies.* A reliable method must generalise across a wide range of anomalies to be effective. A few realistic examples are insufficient to ensure robustness against arbitrary cases.

We address (1) and (2) by learning a distribution for anomalous windows, which enables automatic adaptation and estimation of anomaly difficulty. To reduce bias from the original data, we generate pseudo-anomalies directly in the contextualised space. We learn and model their distribution, $p(\tilde{\mathbf{C}})$, based on the estimated density of normal windows $p(\mathbf{C})$. To satisfy (3), we use a Variational Auto-Encoder (VAE) [32], which provides a simple latent space and inherent diversity via their auxiliary variable $\epsilon \sim \mathcal{N}(0, I)$. Our approach models $p(\mathbf{C})$ with an auto-encoding strategy (Section 4.3) and uses its latent space to generate pseudo-anomalies (Section 4.3). A detailed overview is presented in Figure 2.

Modelling $p(\mathbf{C})$. The core of VAE passes through a latent variable $\mathbf{Z}$ and two distribution estimations $q_\phi(\mathbf{Z} \mid \mathbf{C})$ and $p_\theta(\mathbf{C})$, parametrised by ϕ and θ respectively. By defining $p_\theta(\mathbf{C})$ with its marginal over $\mathbf{Z}$, $\int p_\theta(\mathbf{C}, \mathbf{Z}) d\mathbf{Z}$, we can write the relation between $\log p_\theta(\mathbf{C})$ and the *Evidence Lower Bound* (ELBO) as:

$$\log p_\theta(\mathbf{C}) \geq \mathbb{E}_{q_\phi(\mathbf{Z} \mid \mathbf{C})} [\log p_\theta(\mathbf{C} \mid \mathbf{Z})] - \mathfrak{D}_{\text{KL}}(q_\phi(\mathbf{Z} \mid \mathbf{C}) \parallel p(\mathbf{Z})) \quad (5)$$

$$\geq \text{ELBO}(\theta, \phi; \mathbf{C}). \quad (6)$$

Maximising the ELBO with respect to the parameters θ and ϕ yields a tractable optimisation objective that approximates the true maximum likelihood estimation of the underlying data distribution $p(\mathbf{C})$. We thus obtain the optimisation objective for the modelling of $p(\mathbf{C})$:

$$\mathfrak{L}_{\text{ELBO}} := -\mathbb{E}_{q_\phi(\mathbf{Z} \mid \mathbf{C})} [\log p_\theta(\mathbf{C} \mid \mathbf{Z})] + \mathfrak{D}_{\text{KL}}(q_\phi(\mathbf{Z} \mid \mathbf{C}) \parallel p(\mathbf{Z})). \quad (7)$$

A key modelling choice in VAEs is to consider $p_\theta(\mathbf{C} \mid \mathbf{Z}) = \mathcal{N}(\mathbf{C}; \mathbf{g}_\theta(\mathbf{Z}), \mathbf{I})$, where $\mathbf{g}_\theta : \mathbb{R}^M \rightarrow \mathbb{R}^M$ is the *reconstruction* function. This assumption allows to

reduce the previous equation to:

$$\mathfrak{L}_{\text{ELBO}} \equiv \mathbb{E}_{q_\phi(\mathbf{Z}|\mathbf{C})}[\|\mathbf{C} - \mathbf{g}_\theta(\mathbf{Z})\|^2] + \mathfrak{D}_{\text{KL}}(q_\phi(\mathbf{Z} | \mathbf{C}) \| p(\mathbf{Z})). \quad (8)$$

Finally, the prior $p(\mathbf{Z})$ is also chosen to be simple, $p(\mathbf{Z}) = \mathcal{N}(0, \mathbf{I})$; defining $\hat{\mathbf{C}} = \mathbf{g}_\theta(\mathbf{Z})$ as the reconstructed window, we have the second objective of our entire model:

$$\mathfrak{L}_{\text{ELBO}}(\mathbf{C}_t) = \|\mathbf{C}_t - \hat{\mathbf{C}}_t\|^2 + \frac{1}{2L} \sum_{l=1}^L \sum_{k=1}^M [\mu_l^2 + \sigma_l^2 - \log \sigma_l^2 - 1]_k, \quad (9)$$

where $[\cdot]_k$ denotes the value at the k^{th} dimension of the vector. Note that we estimate the couple of parameters (μ, σ) for each contextualised time-step in the window. Thus, we consider a set of parameters $\mathcal{P}_t = \{(\mu_l, \sigma_l)\}_{l=1}^L$ for each window, which, on average, should follow the same distribution as the prior $p(\mathbf{Z})$. We refer to this part of the network as P_n in Figure 1.

Anomalies from $q_\phi(\mathbf{Z} | \mathbf{C})$. While the first component of the perturbator, P_n , models the distribution of normal windows $p(\mathbf{C})$, we introduce a parallel component P_a to infer the anomalous distribution $p(\tilde{\mathbf{C}})$. Since these distributions are correlated, instead of directly estimating $p(\tilde{\mathbf{C}} | \mathbf{C})$, we leverage the learned latent estimator $q_\phi(\mathbf{Z} | \mathbf{C})$ and model $p_\psi(\tilde{\mathbf{C}} | \mathbf{Z})$, parametrised by ψ . Sampling from $q_\phi(\mathbf{Z} | \mathbf{C})$ naturally induces diversity in the generated anomalies.

As no anomalous examples are available, $p_\psi(\tilde{\mathbf{C}} | \mathbf{Z})$ cannot be learnt generatively like $\mathbf{g}_\theta$. Instead, inspired by adversarial learning [21], we optimise P_a through the classification objective using a reverse gradient, $\nabla_{P_a} = -\frac{\partial \mathfrak{L}_{CE}}{\partial \tilde{\mathbf{C}}_t}$. In this case, however, the learnt distribution $p(\tilde{\mathbf{C}} | \mathbf{Z})$ is approximated only locally around the decision boundary. The gradient ∇_{P_a} vanishes for samples already confidently classified as anomalous, but these distant anomalies are trivial and do not contribute meaningful information; hence, we consider this acceptable.

Architecture. The perturbator is built around the Transformer architecture with an innovative configuration. The encoder q_ϕ parametrises a latent distribution of $\mathbf{C}$; the decoders ($\mathbf{g}_\theta$ and p_ψ) are conditioned on the resulting sampled latent window $\mathbf{Z}^{(i)}$ via cross-attention. To ensure that information is conveyed through the latent $\mathbf{Z}$, *i.e.*, the reconstruction and generation processes depend on the latent representation, we use positional encodings alone as the decoders inputs. Full architectural details and implementation specifics are provided in the supplementary material.

5 Experiments

In this section, we describe the datasets used in our experiments, outline the baseline methods for comparison. We then report the experimental results and

conclude with an ablation study that evaluates the contribution of each component of our approach.

5.1 Datasets

We experimented with four popular multi-variate time-series anomaly detection datasets to validate the effectiveness of ASTER. The datasets encompass different domains; specifically, they are: 1) Pooled Server Metrics (PSM) [1], 2) PUMP [18], 3) Secure Water Treatment (SWaT) [45], and 4) Controlled Anomalies Time Series (CATSv2) [20]. The statistics of the datasets can be found in supplementary. The curated version of the CATSv2 dataset in the TAB benchmark contains 4 subsets, and we train and test on each one. The final results are presented by averaging the metrics for all the subsets.

5.2 Experimental Setup

Metrics. We evaluate our method using five metrics: one threshold-based metric, the F_1 -score, and four rank-based metrics, AUROC, AUPR, VUS-AUROC, and VUS-AUPR. The Volume Under the Surface (VUS) metrics [47] extend traditional measures into a three-dimensional space by incorporating a temporal buffer size, which accounts for slight early or late detections. While AUROC and AUPR evaluate performance across thresholds, they are sensitive to temporal misalignments and ignore the range-based nature of anomalies. VUS calculates the area under ROC or PR curves across all buffer sizes up to the maximum anomaly length, producing a threshold-independent volume that captures a model’s robustness to noise, shifts and timing variations.

For the F_1 -score, an optimal threshold is needed to convert scores into binary labels. In TAB, ten percentile values from the classifier outputs on train and test sets are assessed, and the percentile that maximises the score is used as the threshold. Point Adjustment (PA)-based metrics, commonly used in TSAD studies, are excluded here due to their tendency to inflate performance and obscure true temporal precision [30,15,31,34].

Baselines. We compared ASTER with 11 baseline methods, which were fairly evaluated under the same protocol in the most recent AD benchmark, TAB [48]. We group them into three different categories based on the main approaches they are using; **Deep Learning-based** methods, DAGMM [71], TimesNet [59], and TranAD [57], large **Pre-Trained Models**, Timer [42], TimesFM [16], Moment [22], and Chronos [3]; **LLM-based** methods, GPT4TS [69], UniTime [41], CALF [40], and LLMMixer [33]. We directly use the results for the baselines from the TAB [48] benchmark without any retraining or inference.

5.3 Results

We present the results of ASTER and the selected baselines in Table 1. Comparing ASTER to previous LLM-based methods, our approach significantly improves upon all metrics, except for the CATSv2 dataset. For instance, on the

Table 1: Comparison of TSAD methods over 4 datasets and 5 metrics. Baseline results are from TAB [48], and our results are computed using the same benchmark. **Best** and second best are bold-faced and underlined, respectively.

Data Metrics	Deep Learning-based			Pre-Trained Models			LLM-based						ASTER
	DAGMM	TimesNet	TranAD	Timer	TimesFM	Moment	Chronos	GPT4TS	UniTime	CALF	LLMMixer		
PSM	F ₁	0.007	0.088	<u>0.403</u>	0.143	0.082	0.117	0.131	0.089	0.097	0.102	0.186	0.512
	AUROC	<u>0.637</u>	0.592	0.631	0.556	-	0.545	0.565	0.580	0.579	0.589	0.593	0.697
	AUPR	<u>0.416</u>	0.391	0.388	0.359	-	0.332	0.370	0.376	0.377	0.370	0.385	0.501
	VUS-ROC	<u>0.608</u>	0.593	0.566	0.541	-	0.540	0.547	0.576	0.579	0.587	0.579	0.631
	VUS-PR	0.404	0.395	<u>0.431</u>	0.350	-	0.329	0.362	0.374	0.377	0.374	0.383	0.452
PUMP	F ₁	0.174	0.017	<u>0.454</u>	0.129	0.037	0.140	0.102	0.014	0.020	0.017	0.081	0.458
	AUROC	0.447	0.485	<u>0.800</u>	0.529	0.409	0.485	0.469	0.414	0.549	0.566	0.398	0.839
	AUPR	<u>0.163</u>	0.121	0.110	0.139	0.117	0.129	0.108	0.101	0.129	0.130	0.104	0.254
	VUS-ROC	0.504	0.630	<u>0.792</u>	0.712	0.568	0.669	0.576	0.551	0.690	0.665	0.545	0.833
	VUS-PR	0.235	0.219	0.243	<u>0.260</u>	0.203	0.240	0.175	0.177	0.224	0.204	0.181	0.290
SWAT	F ₁	0.073	0.077	0.321	0.144	0.099	0.132	<u>0.613</u>	0.034	0.063	0.075	0.098	0.695
	AUROC	0.290	0.288	<u>0.818</u>	0.286	0.242	0.263	0.803	0.224	0.234	-	0.240	0.823
	AUPR	0.082	0.107	0.729	0.119	0.086	0.109	0.410	0.081	0.082	-	0.086	<u>0.542</u>
	VUS-ROC	0.328	0.392	0.699	0.399	0.344	0.370	0.814	0.322	0.335	-	0.342	<u>0.757</u>
	VUS-PR	0.250	0.169	0.520	0.172	0.121	0.161	0.449	0.112	0.115	-	0.120	<u>0.506</u>
CATSv2	F ₁	0.088	0.262	0.126	-	0.129	0.060	0.120	0.089	-	0.138	0.081	<u>0.172</u>
	AUROC	0.619	0.712	0.604	0.541	0.633	-	0.624	0.601	0.334	0.616	0.617	<u>0.665</u>
	AUPR	0.090	0.261	0.129	0.112	0.109	-	0.103	0.094	0.039	<u>0.133</u>	0.085	0.106
	VUS-ROC	0.667	0.772	0.625	0.584	0.705	-	0.705	0.676	0.373	0.689	0.689	<u>0.709</u>
	VUS-PR	0.314	<u>0.265</u>	0.130	0.132	0.125	-	0.112	0.107	0.058	0.145	0.099	0.110

Table 2: ASTER with and without an LLM and different fine-tuning schemes.

LLM	Fine Tuning	PSM			PUMP		
		F ₁	AUROC	AUPR	F ₁	AUROC	AUPR
X	X	0.479	0.669	0.447	0.021	0.460	0.087
GPT-2	X	0.354	0.496	0.389	0.370	0.802	0.199
GPT-2	✓	0.361	0.499	0.417	0.265	0.707	0.146
GPT-2	✓(LoRA)	0.512	0.697	0.501	0.458	0.839	0.254

 Table 3: ASTER and its Transformer-based classifier *vs.* an MLP-based one.

Classifier	PSM			PUMP		
	F ₁	AUROC	AUPR	F ₁	AUROC	AUPR
Transformer	0.512	0.697	0.501	0.458	0.839	0.254
MLP-based	0.497	0.724	0.471	0.016	0.307	0.069

PSM and SWaT datasets, the F₁-score is improved by more than 0.3 points, which demonstrates the effectiveness of our method. Compared to all baselines, on the PSM and PUMP datasets, ASTER marks a new state-of-the-art (SoTA) across all metrics. On the SWAT dataset, our method achieves SoTA results on F₁ and AUROC, and ranks second on the other three metrics. Finally, on the challenging CATSv2 dataset, which contains very few anomalous regions, our method achieves the second best scores on a variety of metrics, following TimesNet [59].

5.4 Ablation Study

We conduct ablation experiments on the PSM and PUMP datasets to evaluate different components of ASTER. For the LLM and fine-tuning, we compare four feature-extractor setups: a linear layer only, a frozen LLM, standard LLM fine-tuning [69], and LoRA fine-tuning (our approach). Using only the linear layer is already competitive, highlighting the effectiveness of the classifier and

perturbator. Adding a frozen LLM improves AUROC on PUMP, demonstrating the benefit of contextualisation. Standard fine-tuning slightly helps PSM but harms PUMP, while LoRA achieves the best results on both datasets, showing its superiority for leveraging LLMs in TSAD (Table 2).

To evaluate the impact of the proposed classifier architecture, we replace it with an MLP while maintaining a constant number of trainable parameters. The MLP-based classifier yields inferior performance, especially on the PUMP dataset, highlighting the importance of modelling the sequential nature of the time series (Table 3).

6 Discussion & Future Work

We analyse the perturbator components P_n and P_a , and discuss the limitations and future directions (figures in supplementary). To ensure VAE non-collapse, we monitored μ_l and σ_l and observed convergence near 0 and 1 with deviations $\sim 10^{-5}$ – 10^{-6} , while the reconstruction loss decreased alongside $\mathfrak{D}_{KL}$, indicating that sufficient information is transmitted. Cosine distances between generated pseudo-anomalies $\tilde{\mathbf{C}}_t$ and normal windows $\mathbf{C}_t$ decrease during training, indicating that pseudo-anomalies become more complex while still being distinguishable by the classifier.

We observed that the generated pseudo-anomalies are sometimes insufficiently diverse; however, they remain closer to real anomalies than to normal data, which aids the classifier in learning meaningful boundaries. Future work may enhance diversity through complementary latent sampling, relaxed prior constraints (*e.g.*, MMD), or distributional constraints on P_a . Even though ASTER converges properly, we also monitored large magnitude differences in its loss components, indicating that the objectives are not being optimised equally and suggesting potential benefits from balancing strategies. Finally, the linear layer used to map time-series to token embeddings may be restrictive, and non-linear alternatives such as MLPs or Kolmogorov–Arnold Networks [43] could manage the complex alignment better.

We note that evaluating on more datasets would better assess ASTER performance, given the limitations and heterogeneity of multivariate TSAD data [60].

7 Conclusion

In this work, we tackle unsupervised TSAD and overcome the limitations of handcrafted, dataset-specific pseudo-anomalies by introducing ASTER, the first framework to generate pseudo-anomalies autonomously in a learnt latent space. Leveraging pre-trained LLM representations, the expressive capacity of Transformers, and the modelling power of VAEs, ASTER produces challenging pseudo-anomalies to adversarially train a robust Transformer-based classifier. Extensive experiments on the TAB benchmark demonstrate that ASTER achieves state-of-the-art performance across datasets, outperforming existing LLM-based methods. Future research could focus on further enhancing the perturbator by

enforcing anomaly diversity through advanced latent sampling, constraints, or more complex decoder inputs (*e.g.*, learnable tokens), exploring the interpretability of the latent space and the generated pseudo-anomalies, and investigating various pre-trained LLMs and their differing representational performance with enhanced time-series-to-tokens translation modules.

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